

Data-Driven Computer Animation

Lecture 1

Introduction

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Teaching Staff

- Lecturer:
 - Taku Komura (me), Professor in Computer Science
- Teaching Assistant:
 - Tianyang Xue
 - Leo Ho



Today's topics

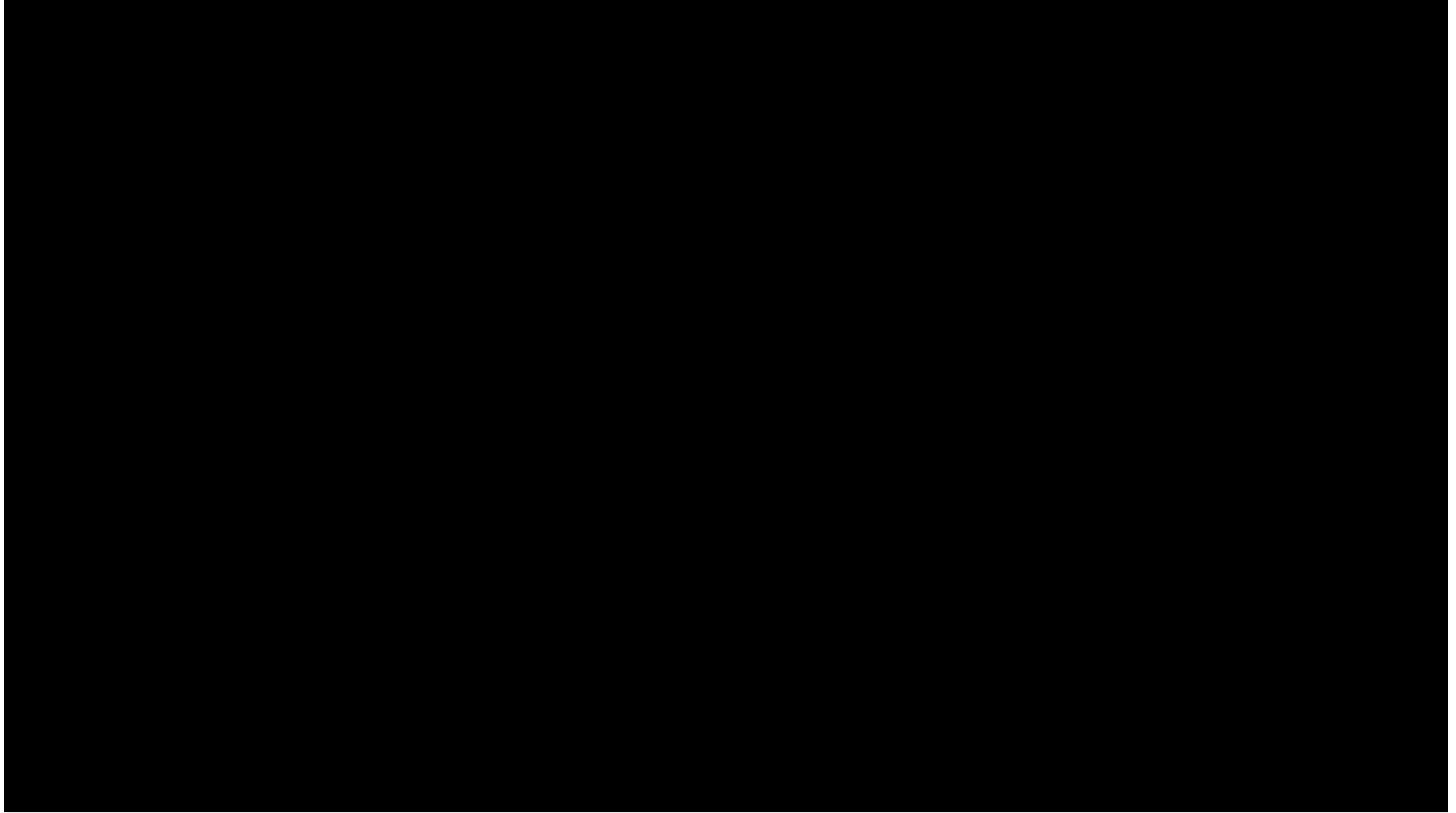
Overview of the lecture

- Introduction to computer animation
- Topics covered in the lecture
- Motivation / Group research
- Other applications of computer animation

What is computer animation?

- Creating moving images via the use of computers.
- Subfield of computer graphics
- Applications :
 - Films & Special Effects
 - TV Programs
 - Computer Games
 - Web Contents
 - VR and AR
 - Prepare training data for machine learning
 - Design of buildings, environment etc.

Demo Animation



Topics of computer animation

- Character animation (3D animation)
 - Keyframe animation, motion capture
 - Skinning, facial animation
 - Motion synthesis by machine learning
 - Crowd simulation
- Physically-based animation
 - Rigid objects
 - Cloth, strands, deformable objects
 - Fluids (fire, water, smoke)
 - Finite Element Method (soft materials)
- Geometry modelling (2D, 3D)
- Computer animation (2D)

Character Animation

- Controlling characters
 - Different body structures (humans / cartoon characters / animals / ...)
 - using real human movements (MoCap)
 - manually creating the movements (KeyFrame)



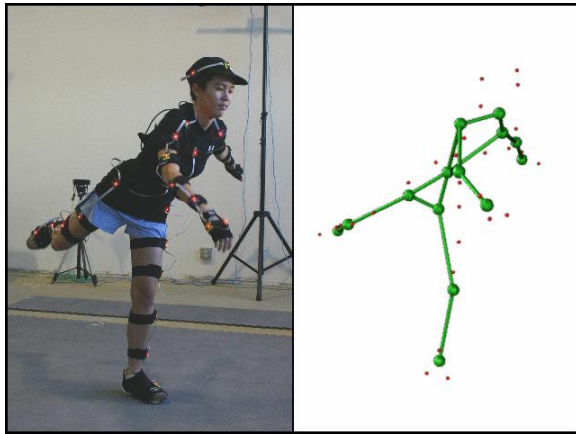
Keyframe Animation

- ❑ The keyframe postures are designed by the animator
- ❑ The inbetween motion is created by interpolation



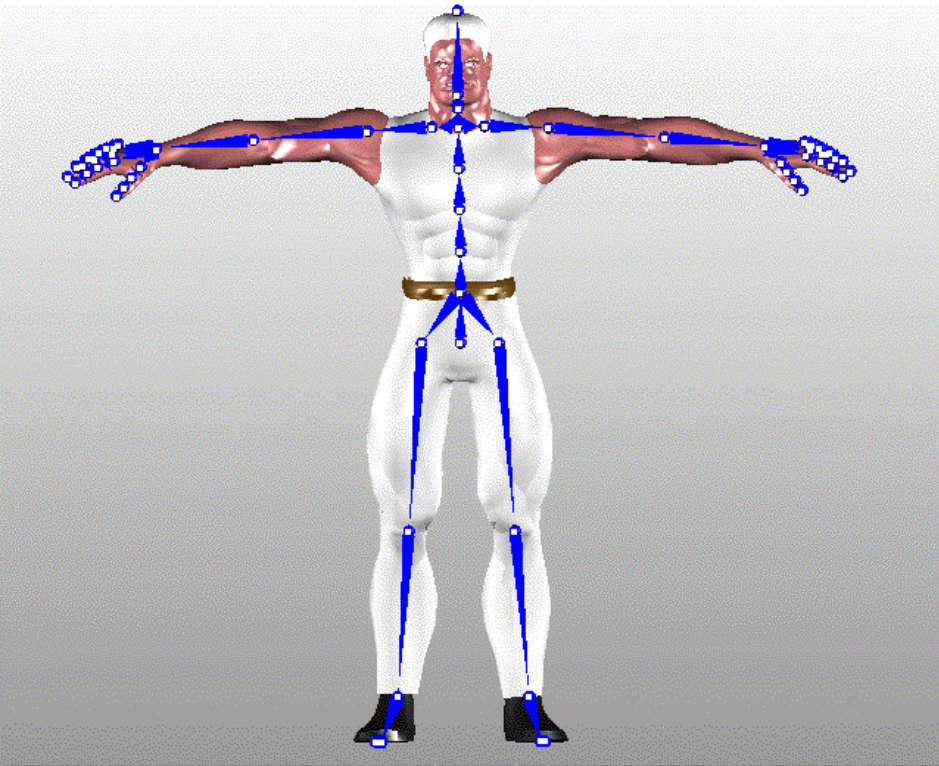
Motion capture

- Digitizing the human movements
- Tracking the movements of the markers
- Apply them to virtual characters



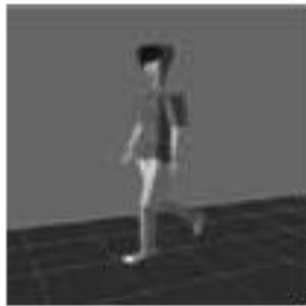
Skinning

- Need to decide how the surface deforms according to the movements of the skeletal bones

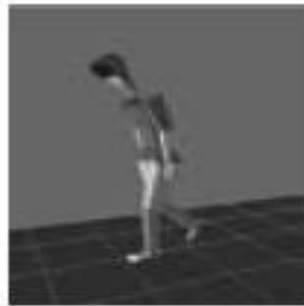


Motion Editing

- Making use of the captured motions for different scenes
- Adjust the movements so that they satisfy constraints



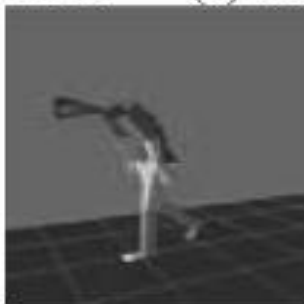
(a) Normal walk



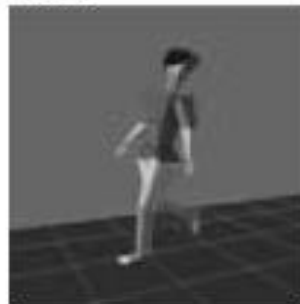
(b) Tired walk



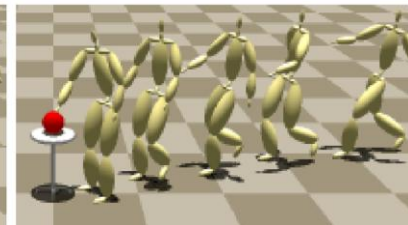
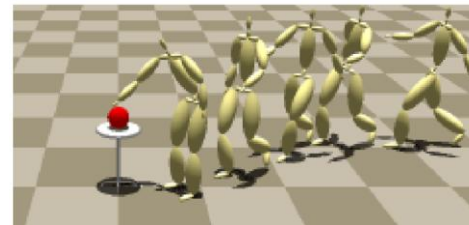
(c) $s = 0.5$



(d) $s = 2.0$

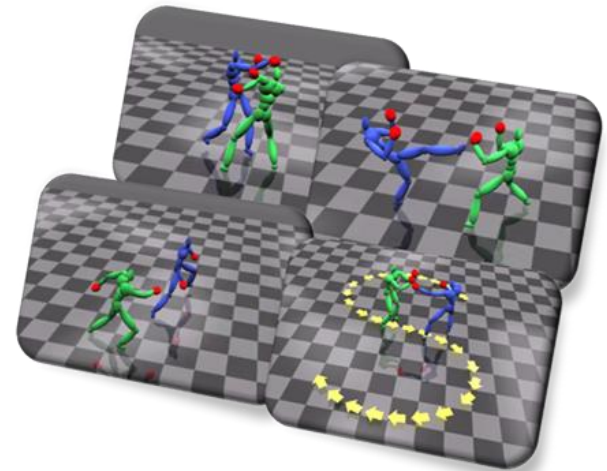
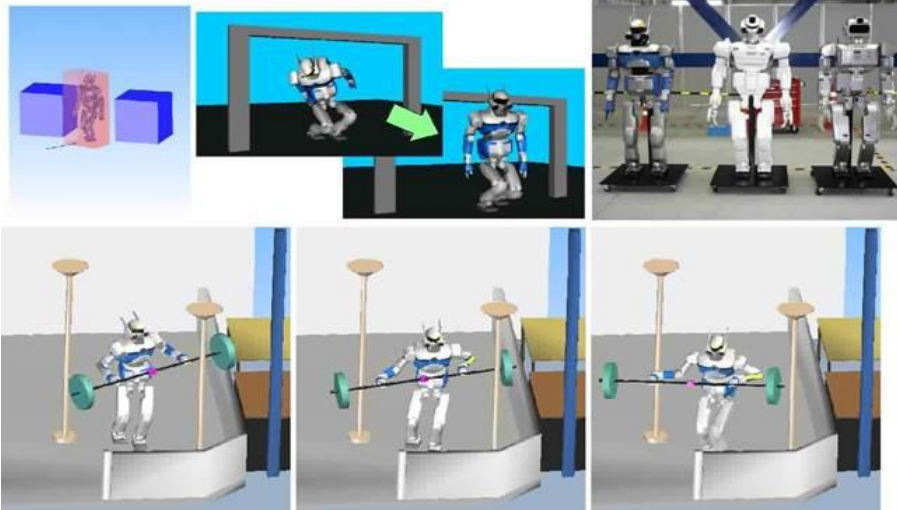
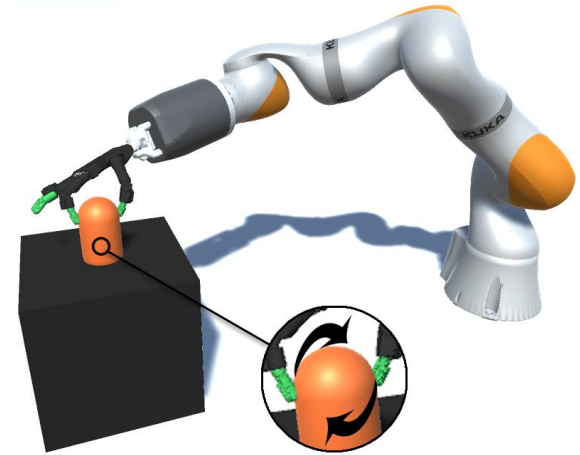


(e) $s = -0.5$



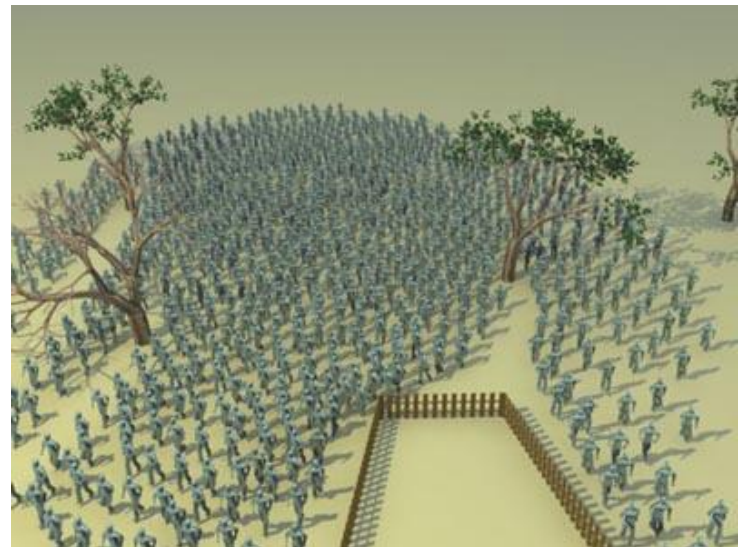
Motion Planning

- How to switch from one posture to another without colliding with other objects / characters
- How to control characters so that they behave smartly



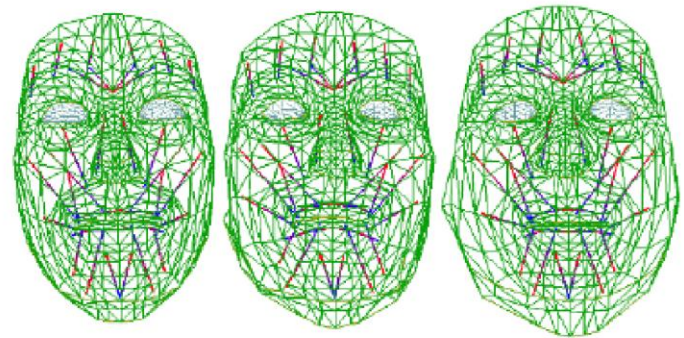
Crowd simulation

- Simulating the pedestrians in the streets, the audience of a concert, soldiers in the battlefield, etc.
- How does one's movement affect those of the others



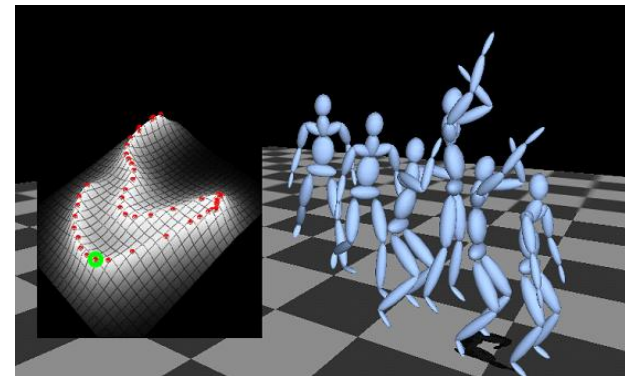
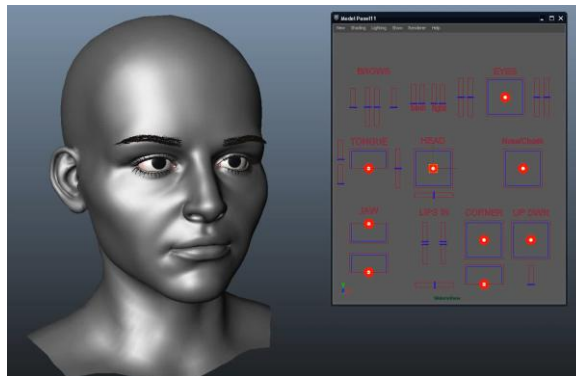
Facial animation

- Animating the face by
 - Motion capture data
 - Using musculoskeletal models

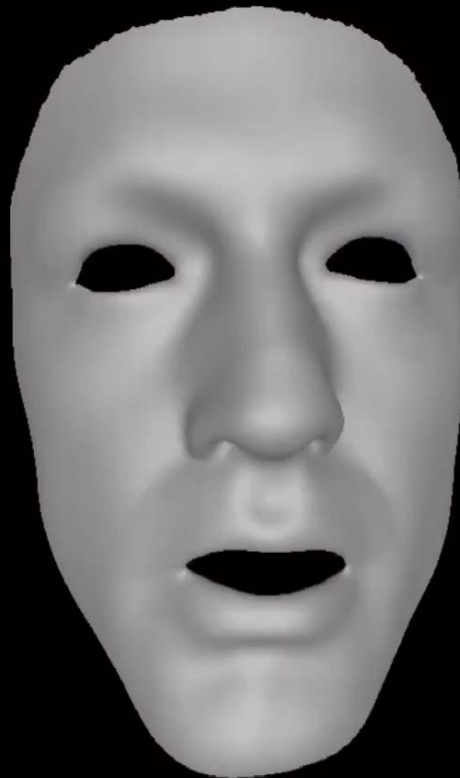


AI/Machine Learning Techniques for Character Animation

- Learn how the humans move using the motion capture data
- Apply them for character motion synthesis
- Controlling high dimensional models (face, skin, high DOF articulated characters) with low dimensional control signals



Face animation from speech



Sad white boy perturbed a woman. Angry blue man and a green guy.

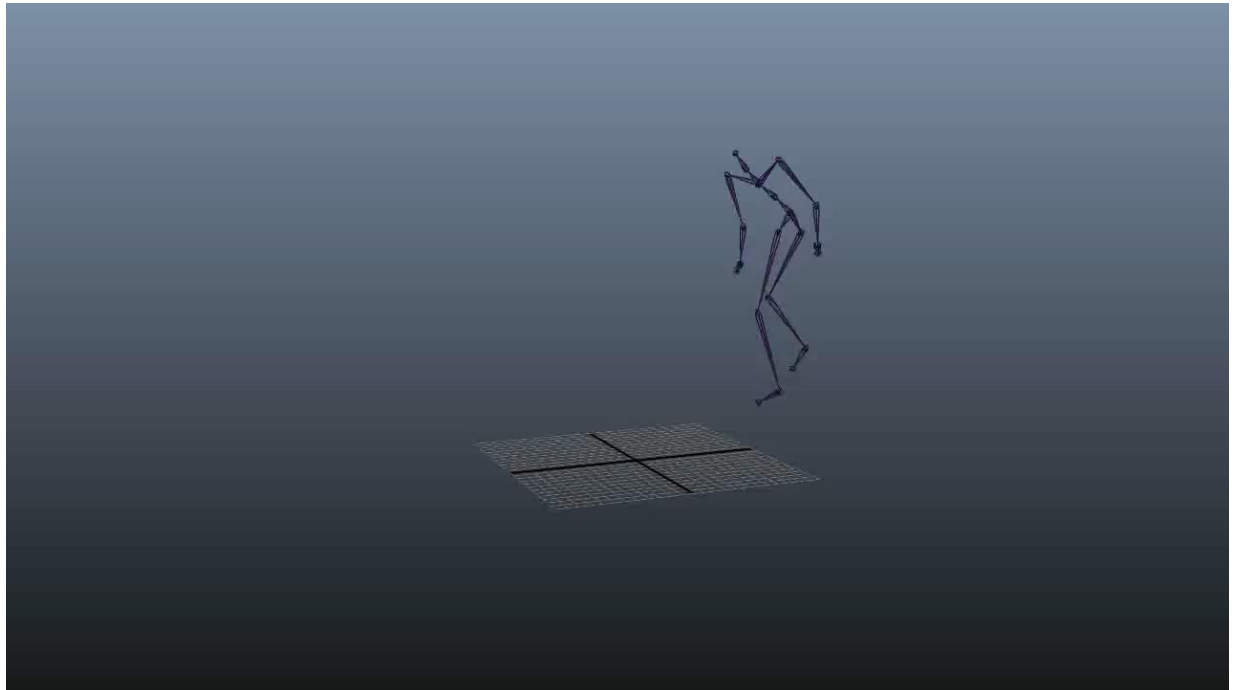
Speech from TED: <https://www.youtube.com/watch?v=04-JxYnzcq0>

Deep Learning Techniques for Character Animation

- Neural network-based time series models are often used nowadays to produce character movements that follow the user instructions.
- Neural networks are useful as they can compress the data a lot. Also relatively fast to compute at runtime.



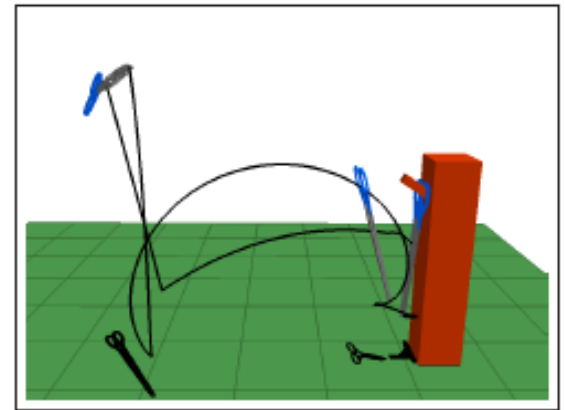
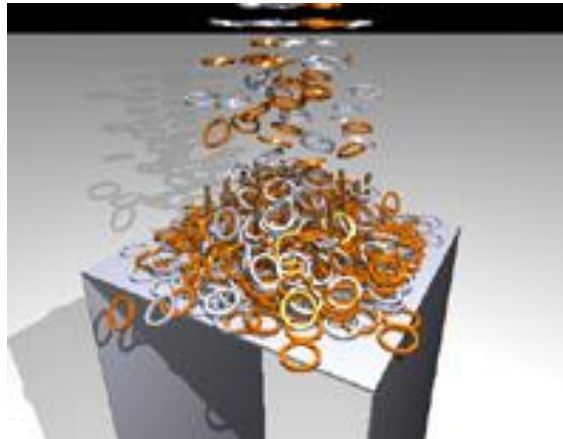
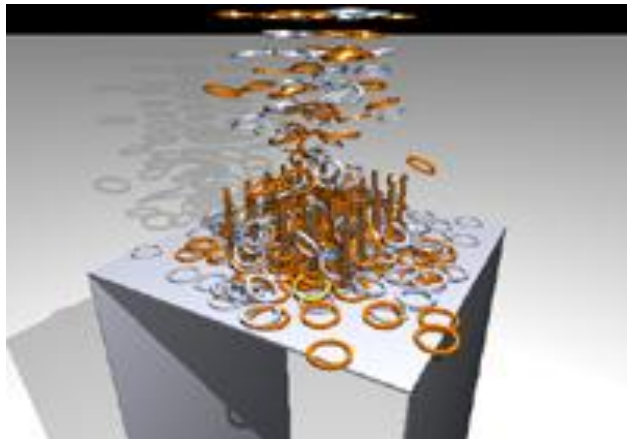
Phase-Functioned Neural Networks

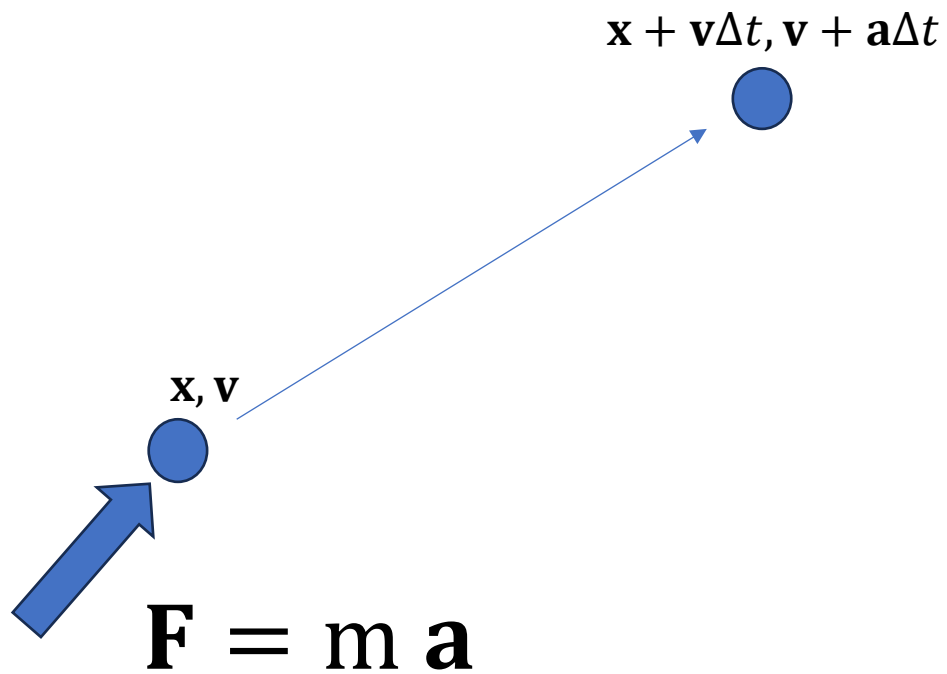




Physically-based Animation: Rigid Objects

- Simulating rigid objects flying, colliding, and bouncing
- Technical issues
 - Collision detection, response
 - Computing the contact forces, friction forces

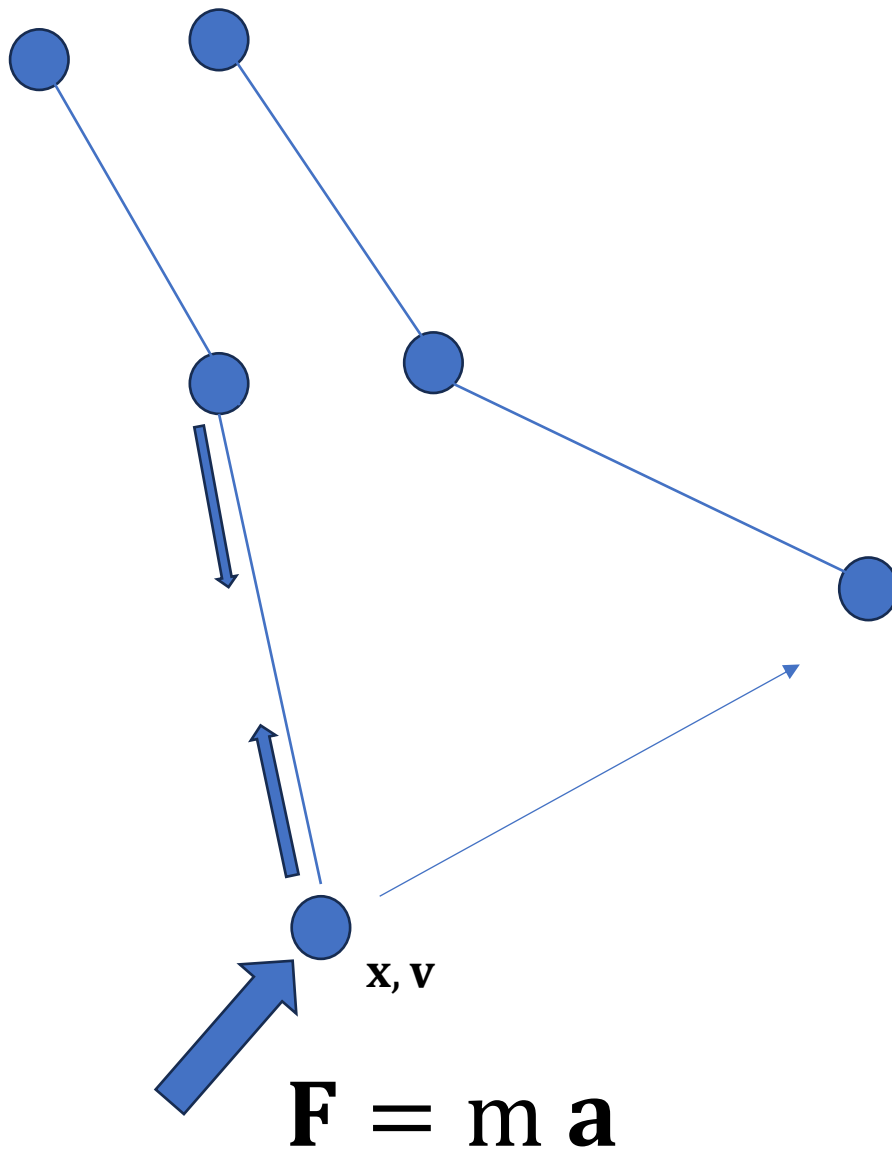




Physically-based animation: Hair

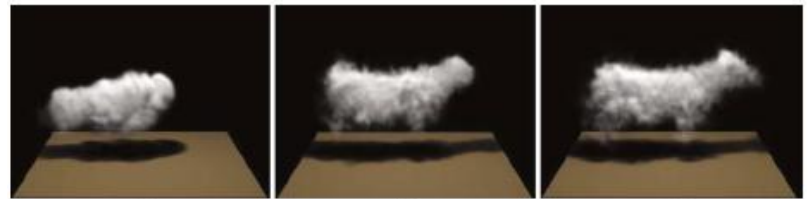
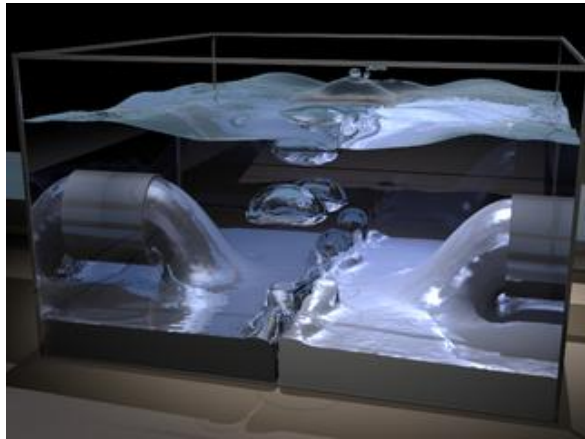
- How the hair moves when the wind blows
- Need to take into account
 - the physical properties of the hair,
 - Collisions between the hair

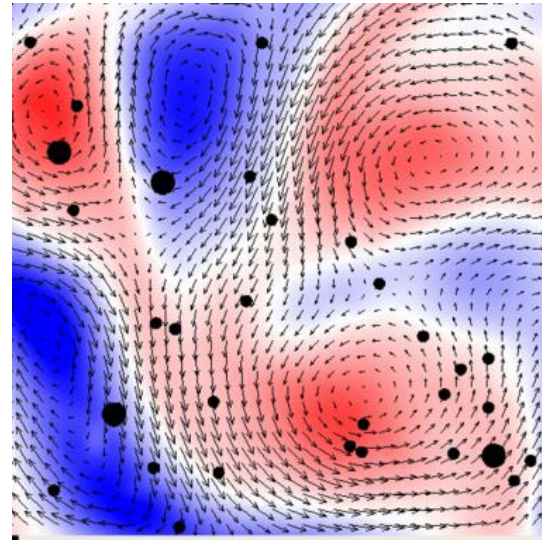
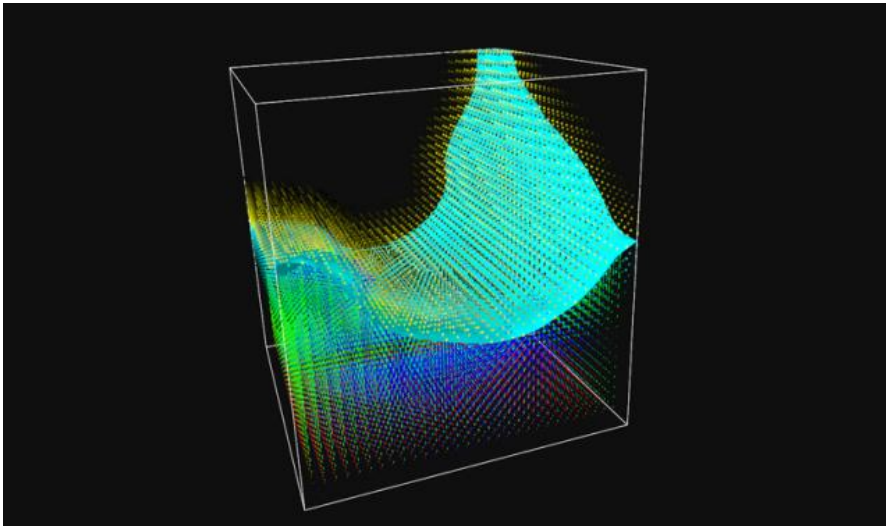
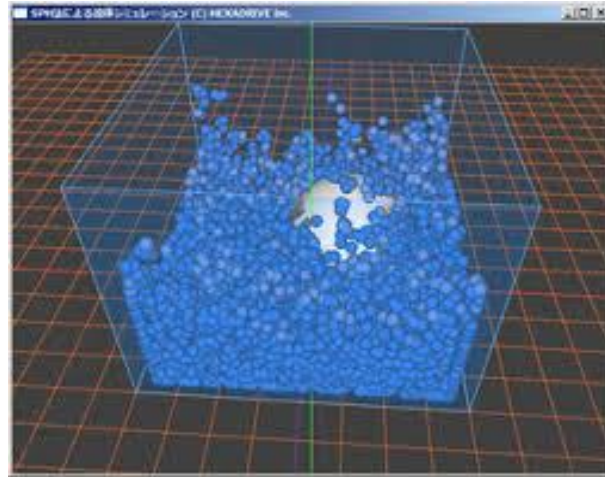
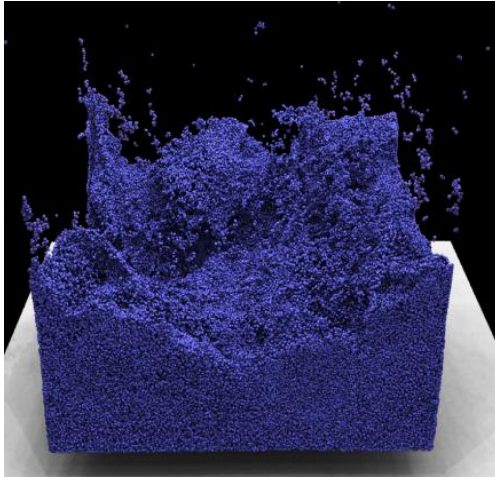




Physically-based animation: Fluids

- Simulating liquid, mud, fire, bubbles
- How to efficiently simulate the motion of the fluids
- How to control the fluids so that the animator can get what s/he wants



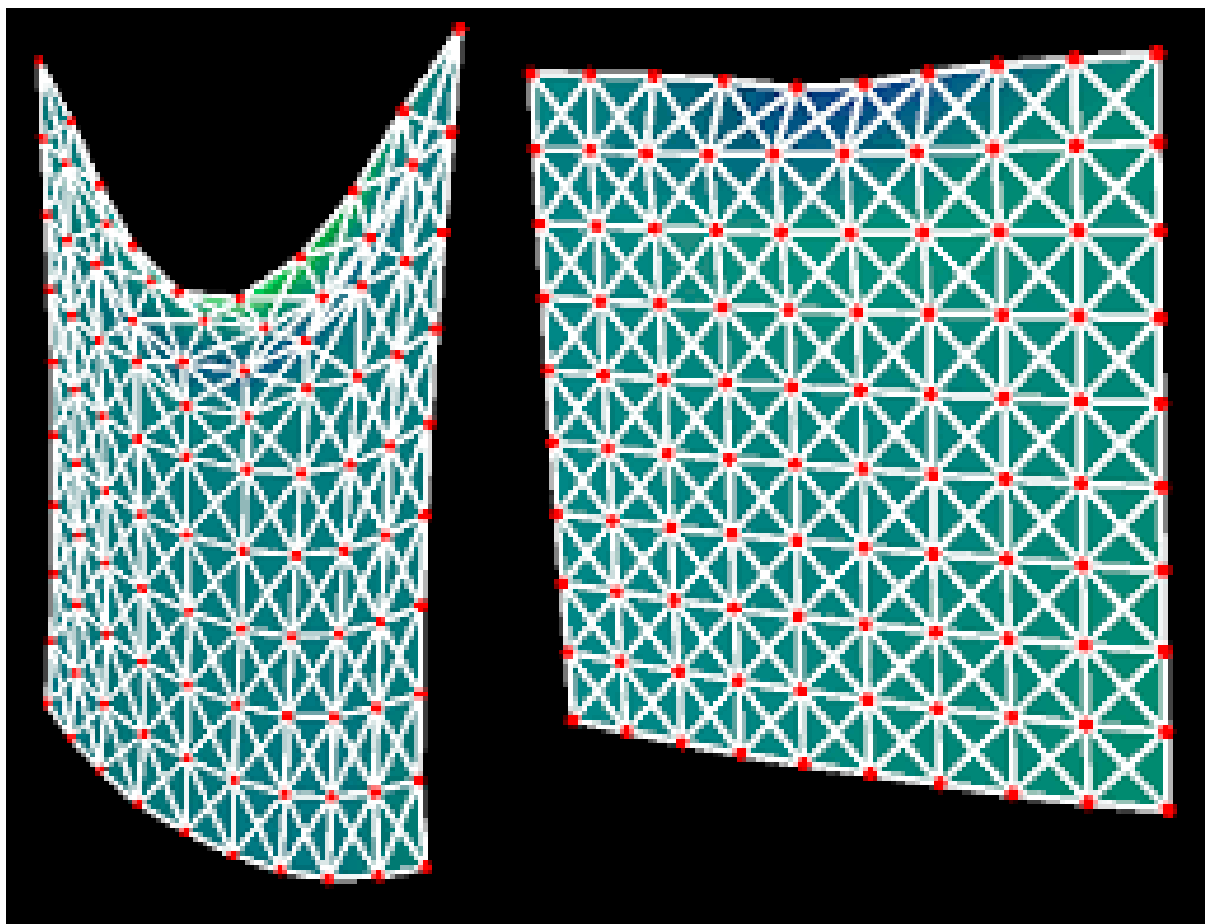


<https://david.li/fluid/>

Physically-based Animation: Cloth simulation

- Simulating the movements of clothes when the body moves
- How the wind affects the deformation of a cloth

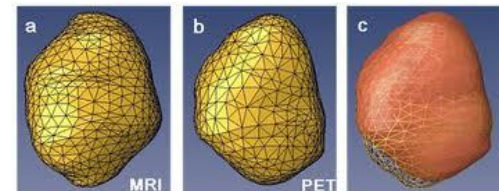
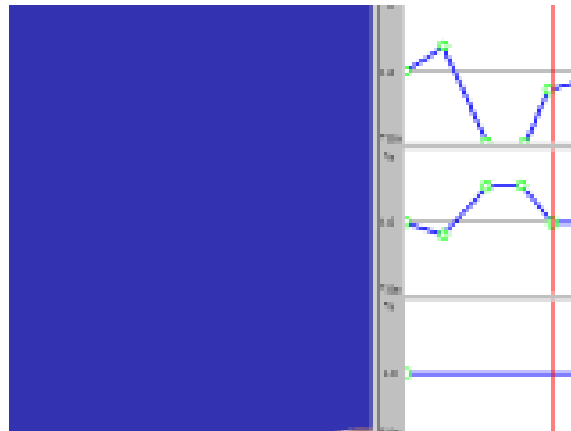
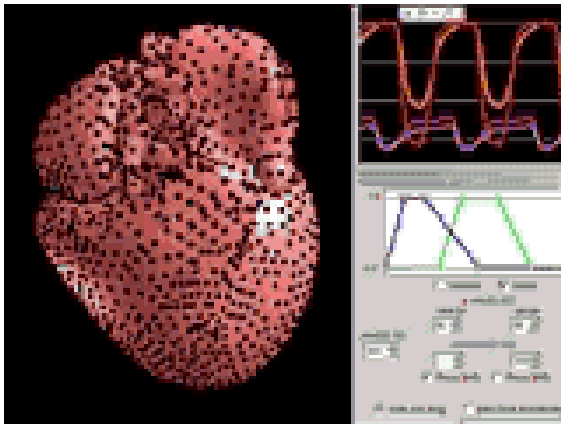




Finite Element Method

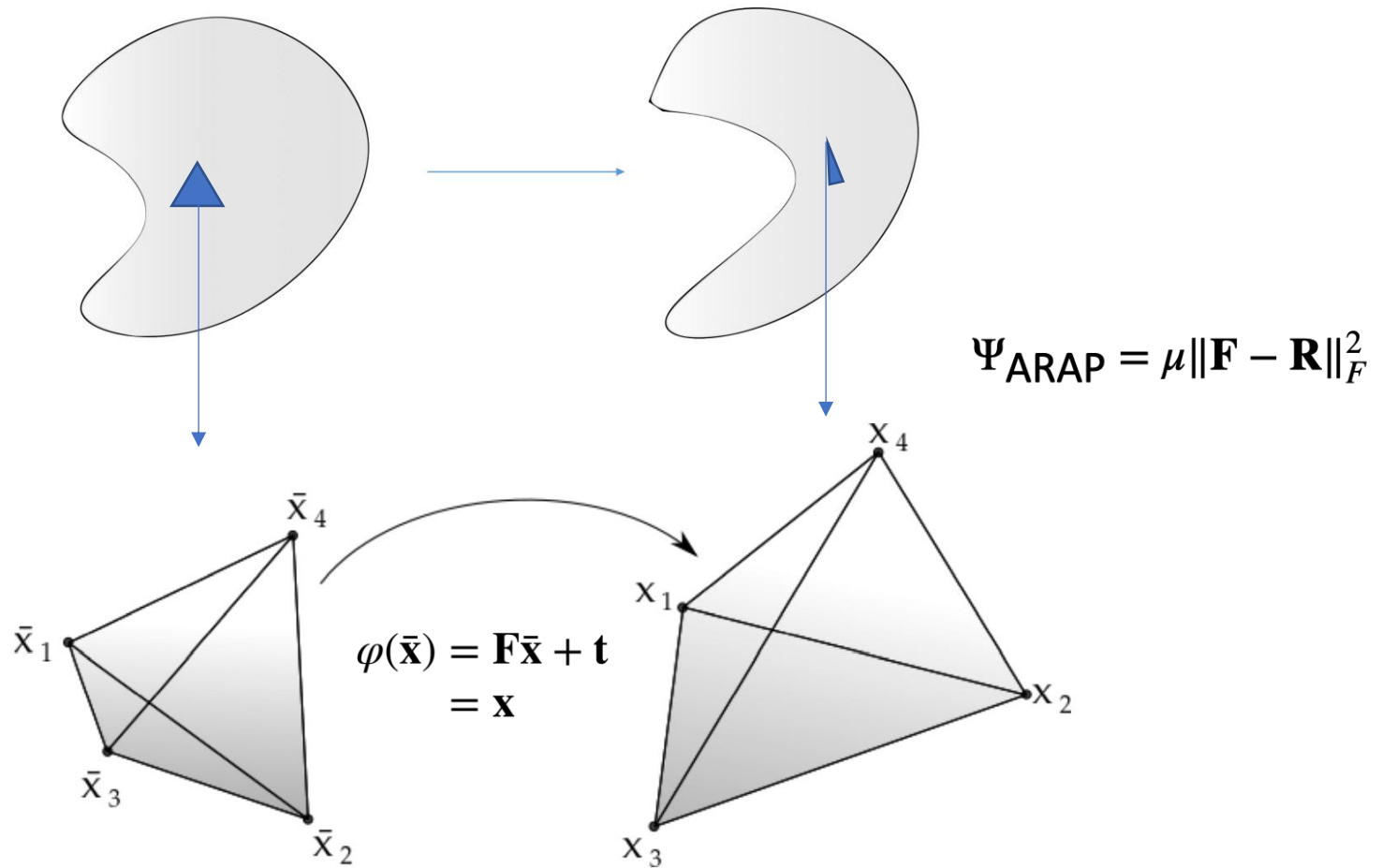
Needed to simulate soft materials like jelly fish, human heart

FEM is also used for the analysis of hard objects like buildings, bridges, aircrafts etc



On courtesy of Dr. Takashi Ijiri

Deformation Gradient



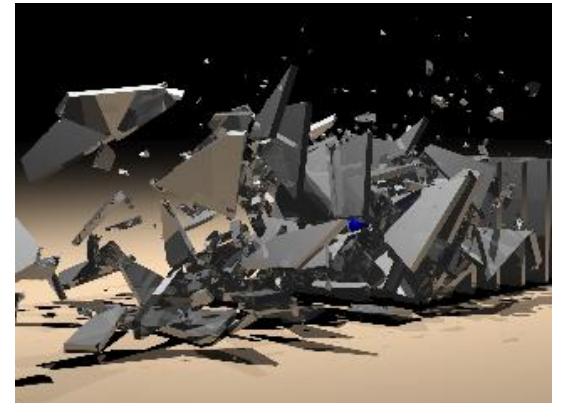
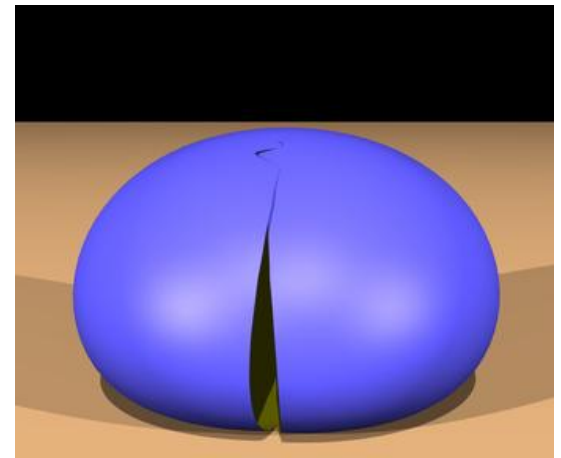
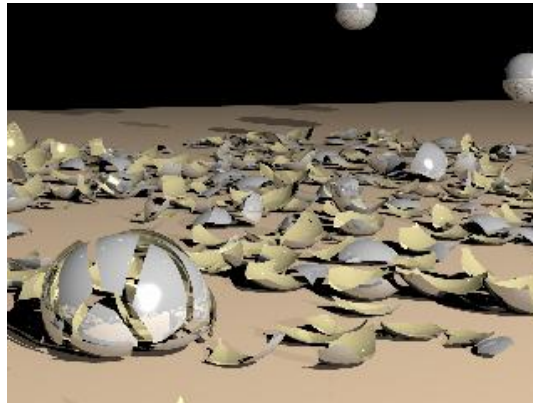
Physical Simulation





Wrecks/Crashes/Destruction

- Simulate how / where the destruction starts and expands





Simulating brittle fracture with material points

Linxu Fan, Floyd M. Chitalu, Taku Komura

Physically-based Animation: Forward Dynamics

- Simulating the body as articulated objects connected at the joints
- Easy to simulate ragdoll motion
- Difficult to simulate voluntary motion



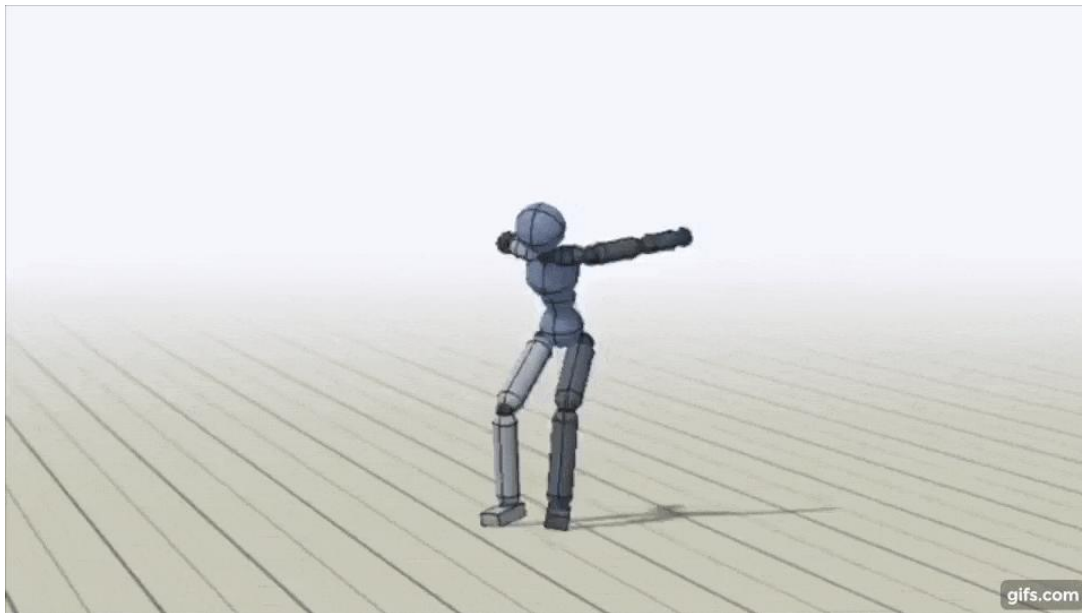
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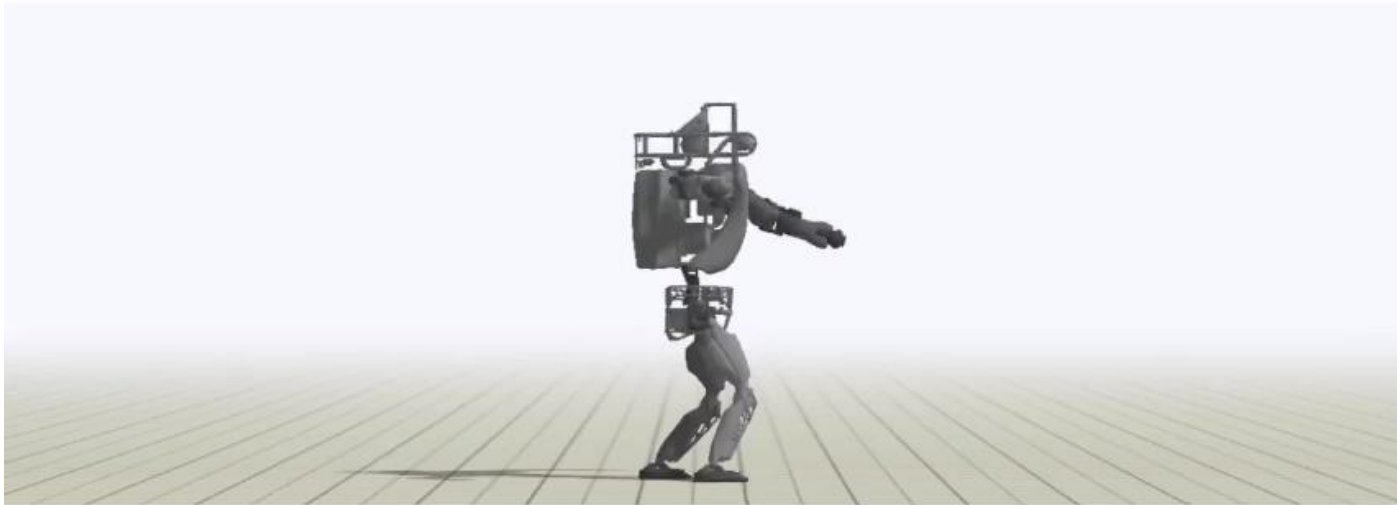
Deep Reinforcement Learning for Tracking Human Motion

- In forward simulation of articulated humans, it is difficult to compute the joint torques to track some motion, especially when the perturbation is very random.
- Researchers propose to use deep reinforcement learning to track a reference motion



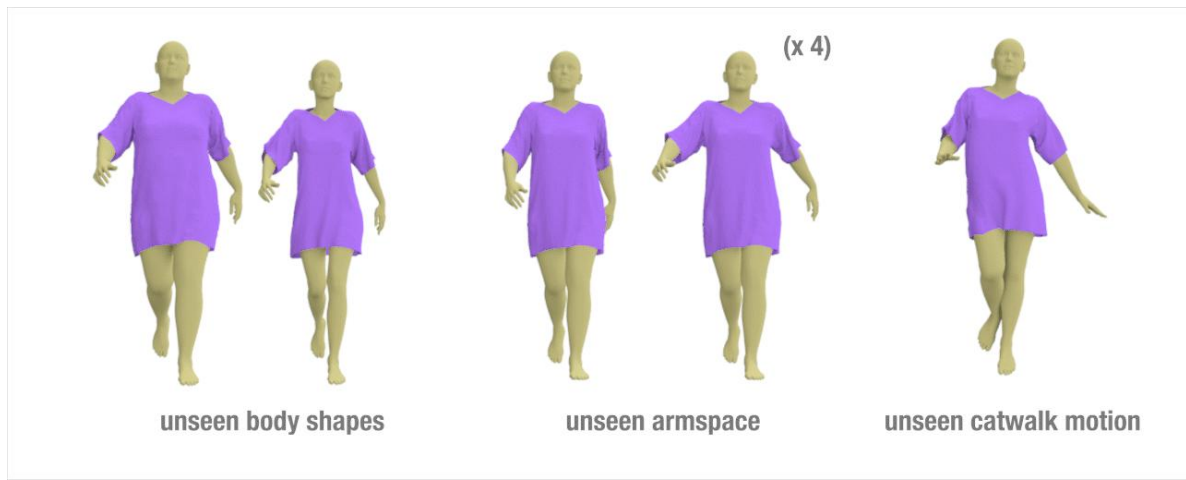
Deep Reinforcement Learning for Tracking Human Motion

Atlas: Spinkick



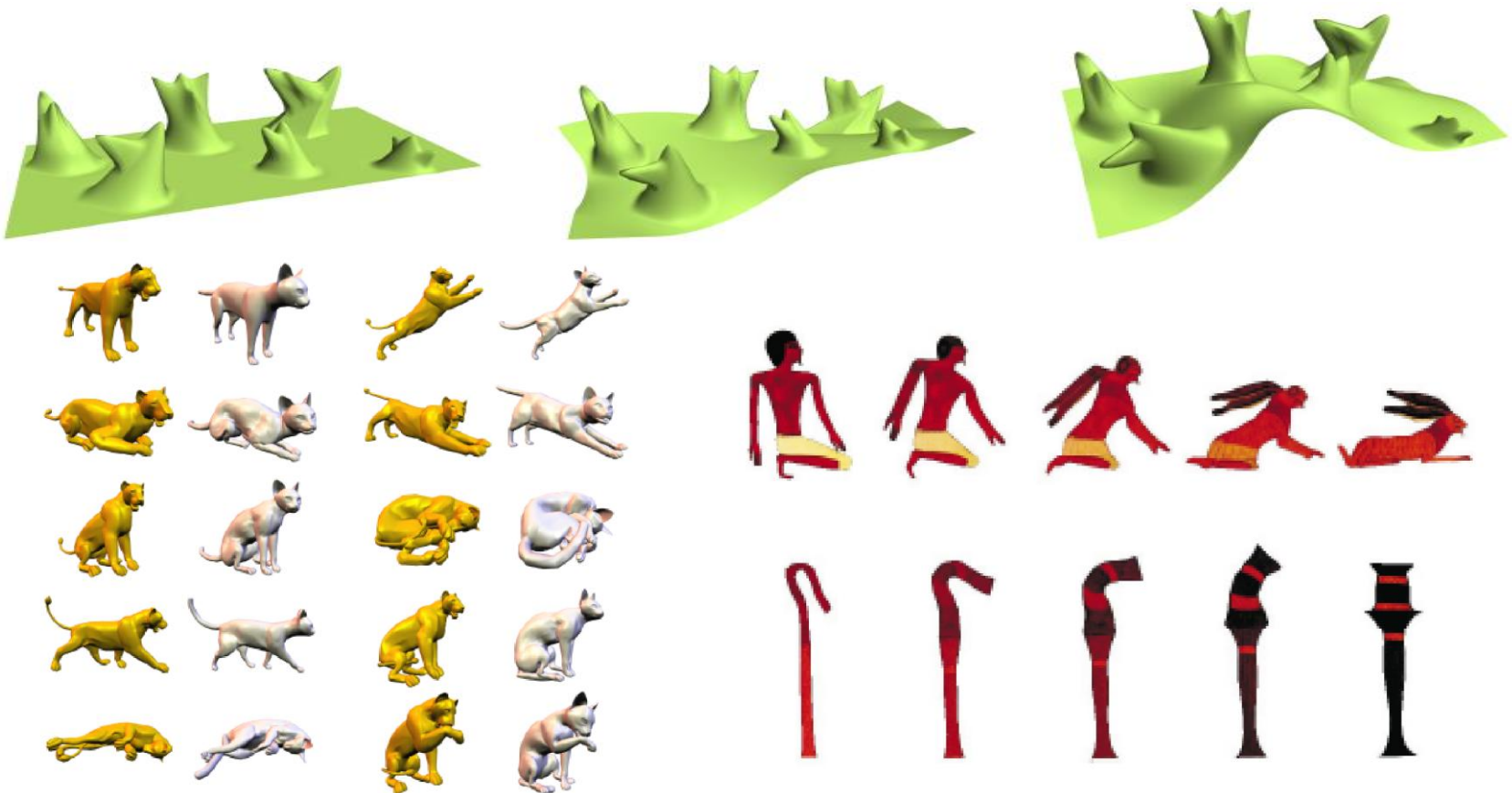
Data Driven Physics Simulation

- Instead of simulating the response of the objects by physics rules, some researchers propose techniques to learn the updates of the state.
- Using the simulated results for supervised learning



Shape Modelling and Editing

Designing shapes, editing shapes



2D Cell animation

- For a 30 minute cartoon, 3000 cell pictures are needed
- This requires a month labour of 50 professional drawers
- How to create 2D Cell animation efficiently
 - Using 3D graphics and render in a 2D cell animation fashion
 - lighting, shadows, deformation



Today's topics

- Introduction to Computer Animation
- Topics covered in this course
- **Motivation / Group Research**
- Other applications of computer animation

Motivation of Research

- Improving the realism of the character movements, simulated scenes
- Improving the efficiency of animation (faster, less memory)



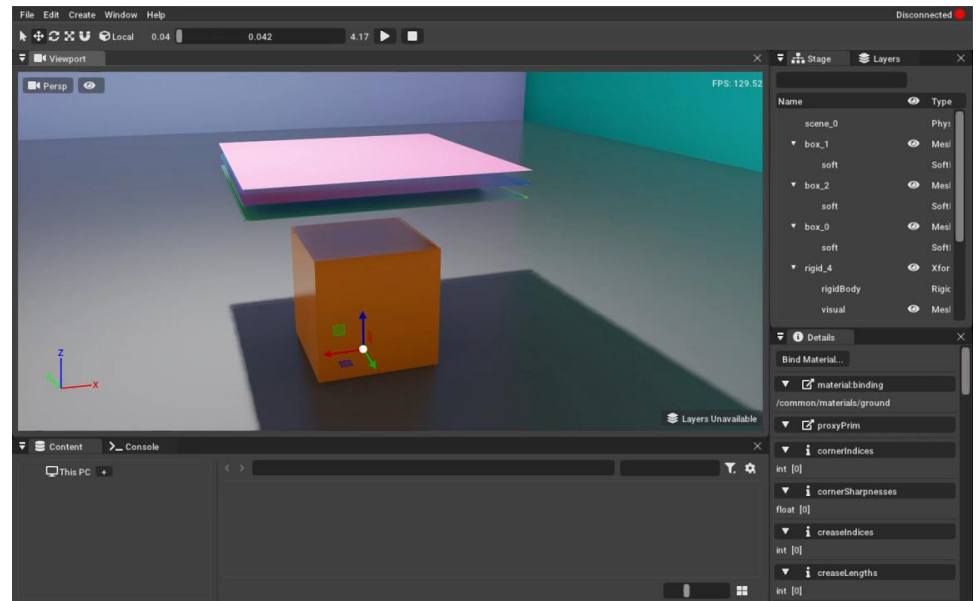
Offline vs. Real-time

- Computer animation techniques are often very expensive
 - Physical simulation may require a lot of memory and mathematical operations
 - Many techniques require optimization with many variables
 - This is fine for applications such as movie production
 - The animators can generate the scene, fix the parameters if they don't like the results and render the results again – can repeat such cycles



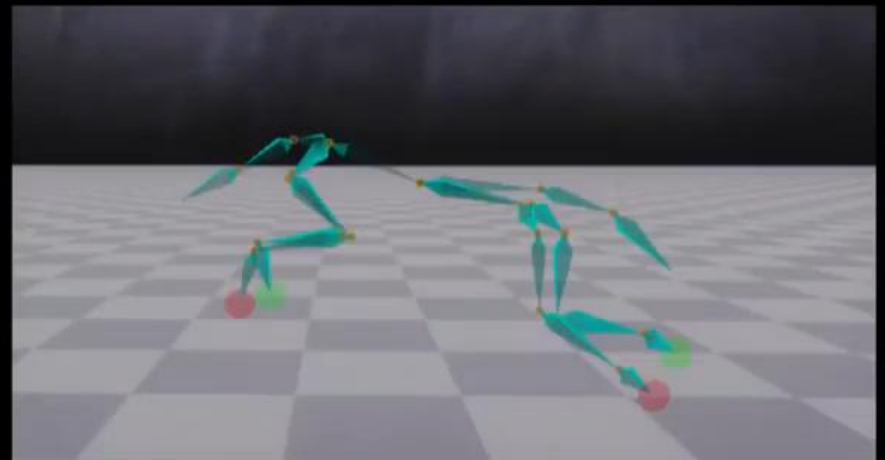
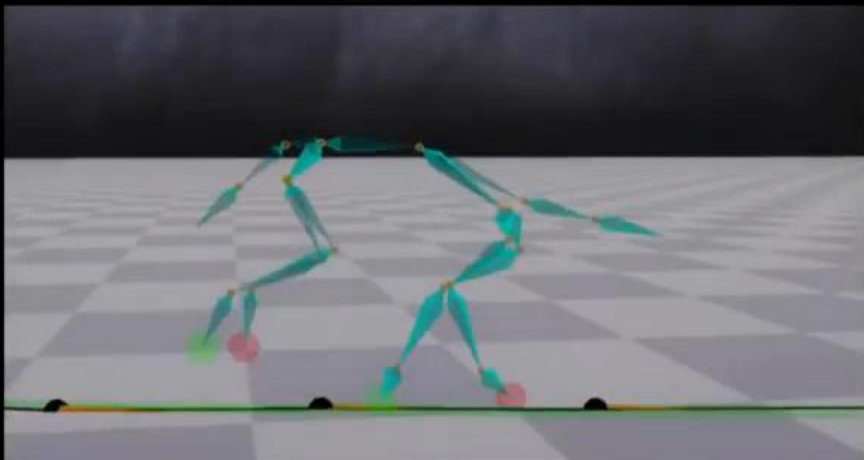
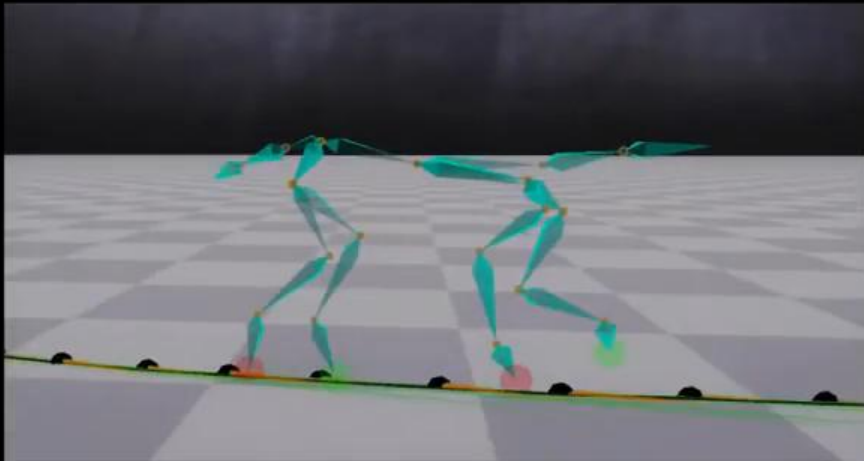
Offline vs. Real-time

- On the other hand, real-time applications such as computer games and virtual reality require quick response (30 frames per second)
- It is often the case that techniques are first developed for offline animation and then improved such that they can run in real-time later.



Mode Adaptive Neural Networks

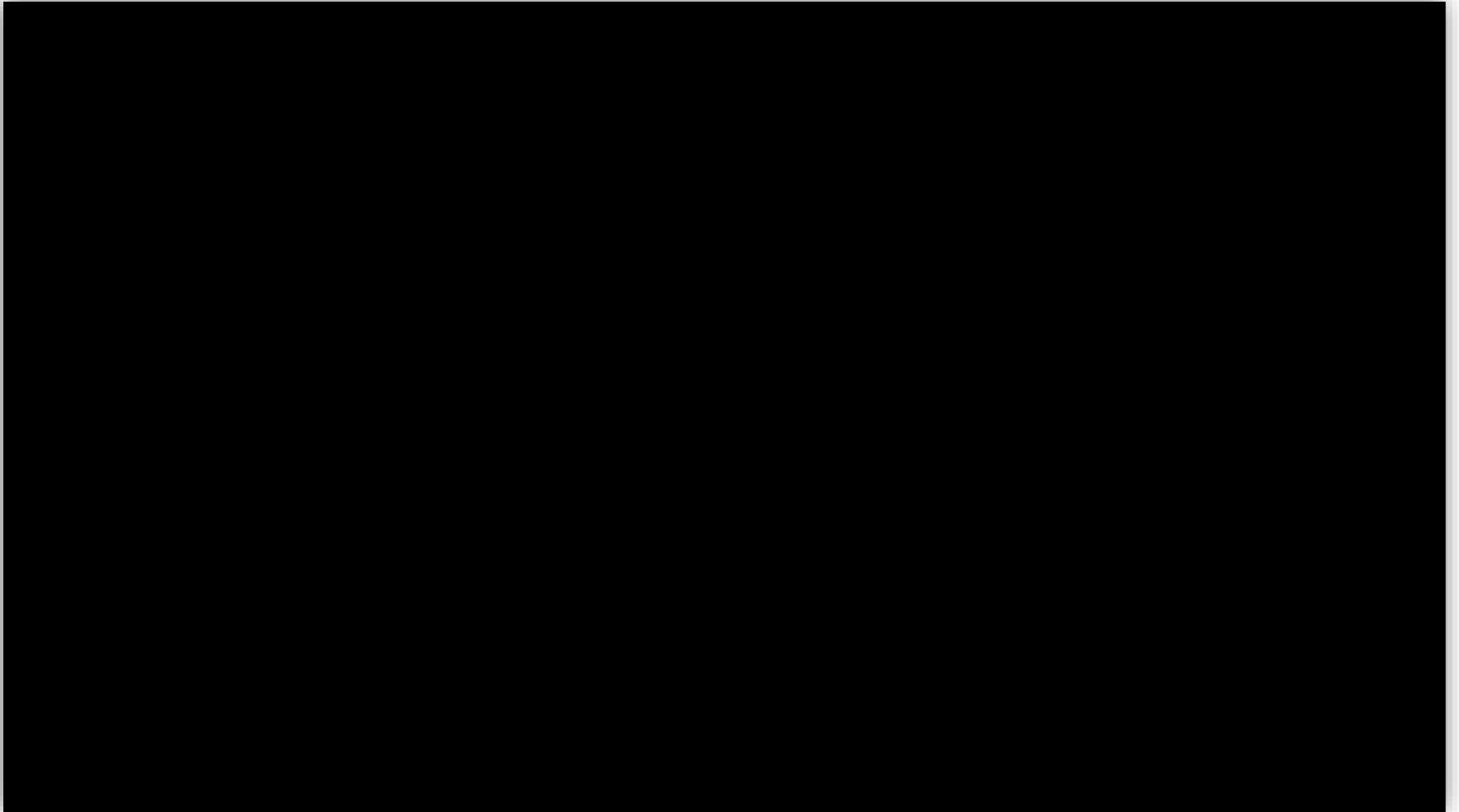
1h Quadraped Motion Capture Data



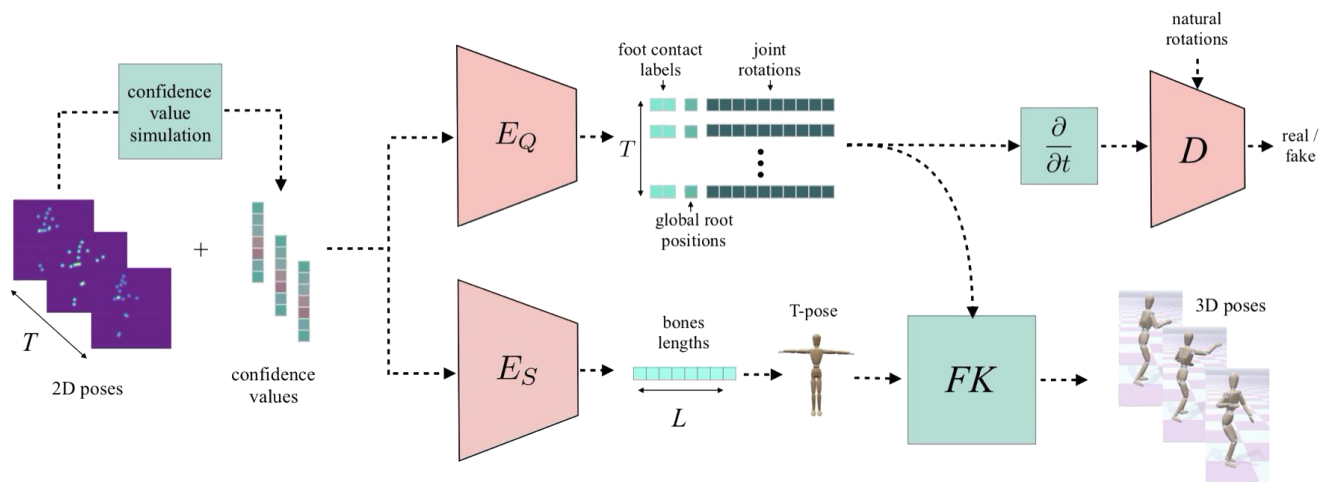
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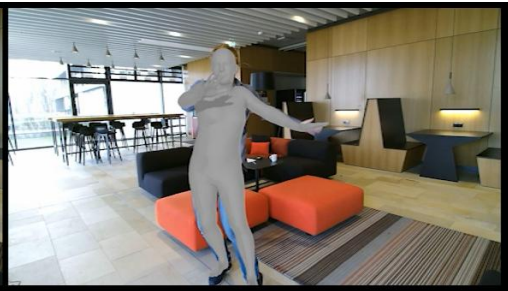
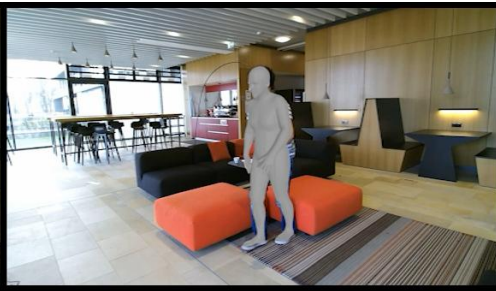
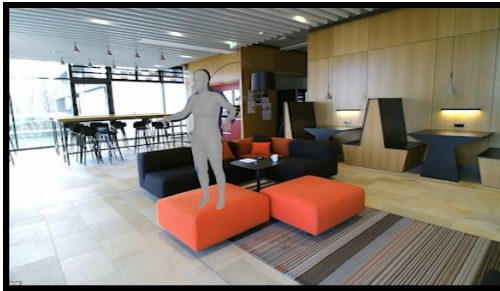
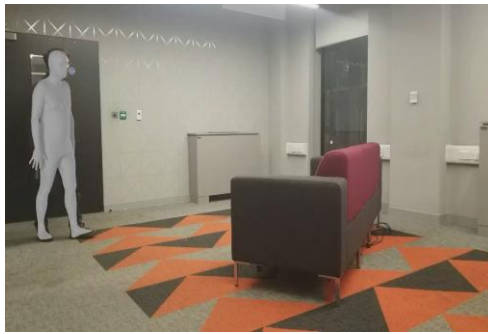


Local Motion Phase

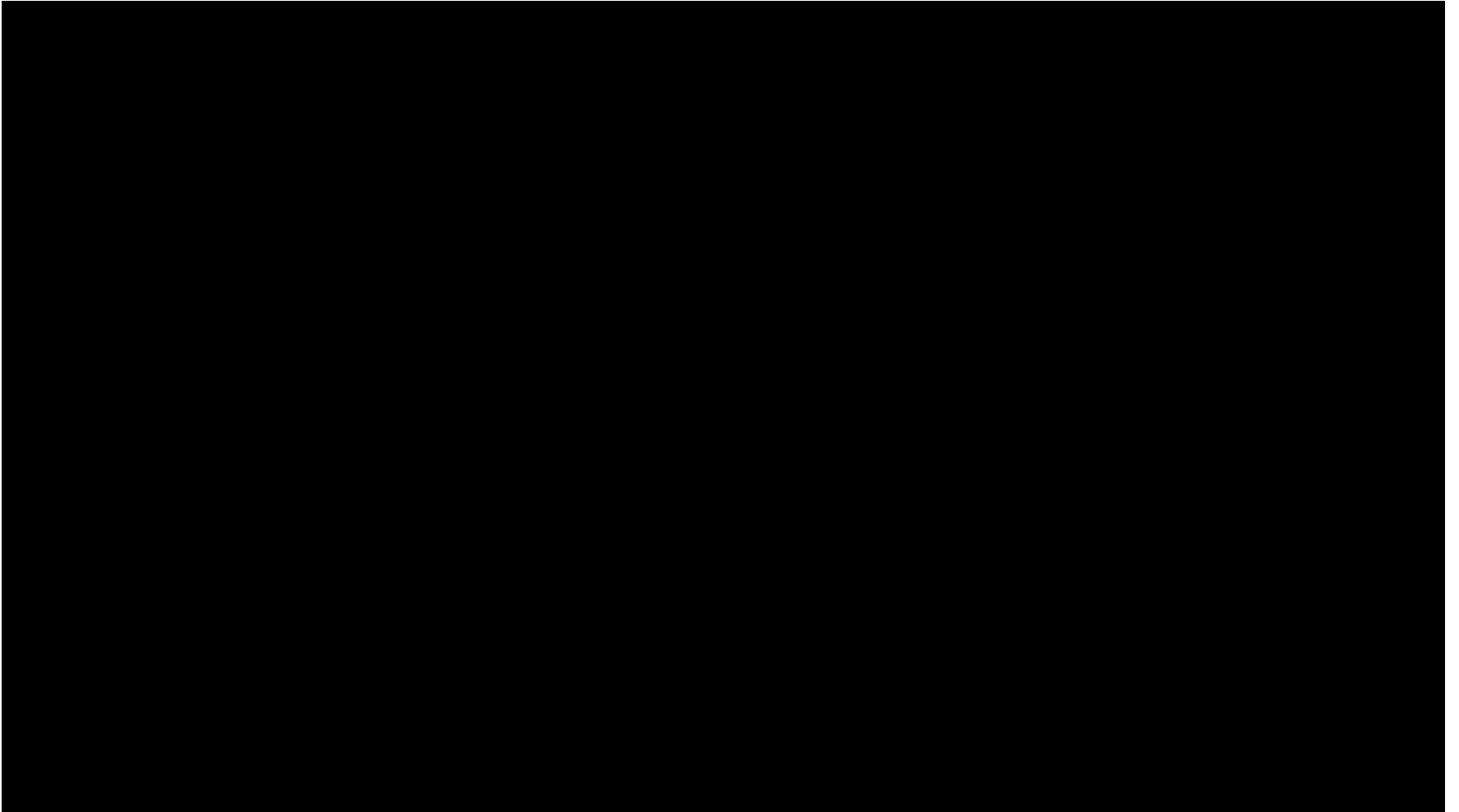


Video-based Motion Capture





Face retargeting



Input

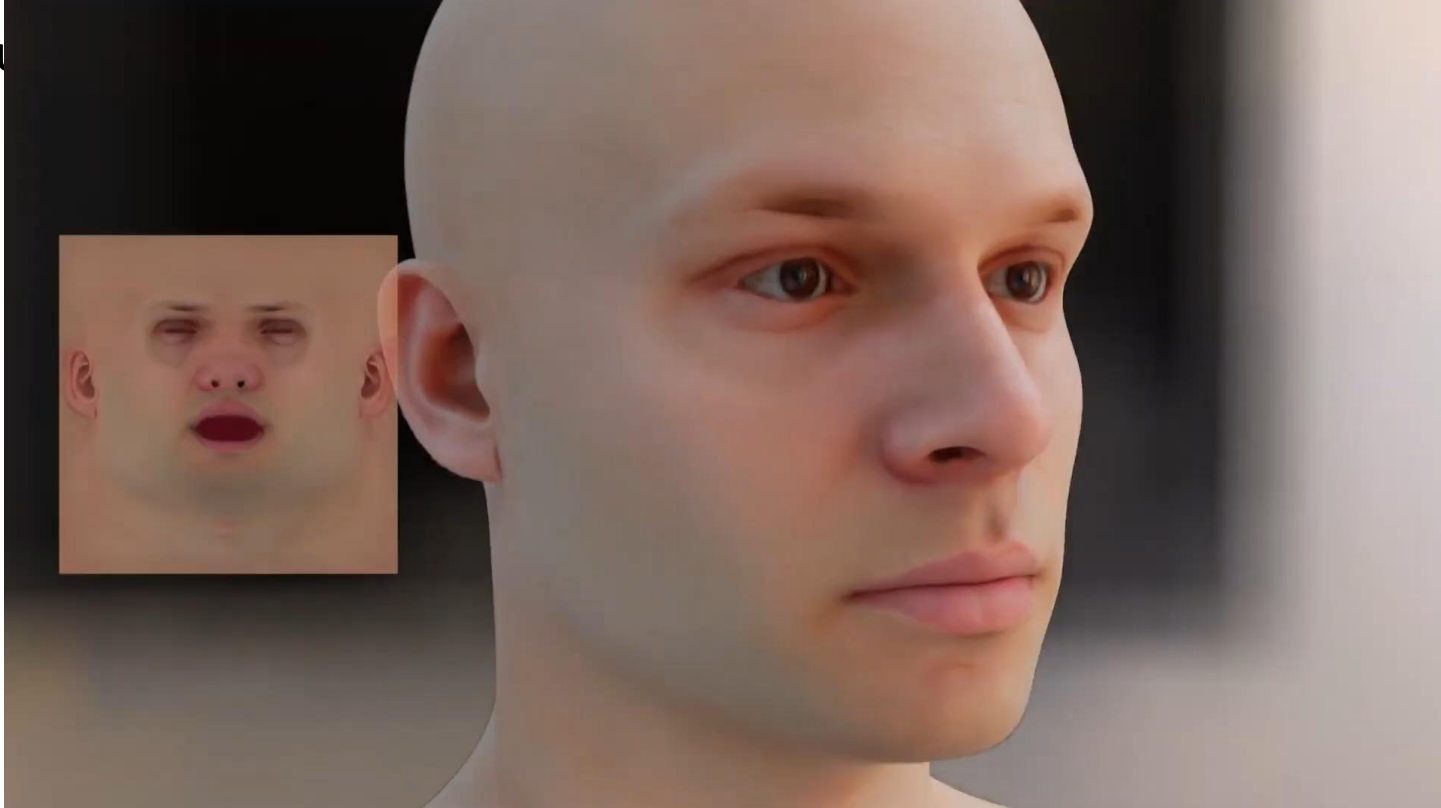
Prompt a middle-aged male with a broad forehead, prominent cheekbones, and deep-set eyes with heavy eyelids.



TransGS

- Result

-
-
-

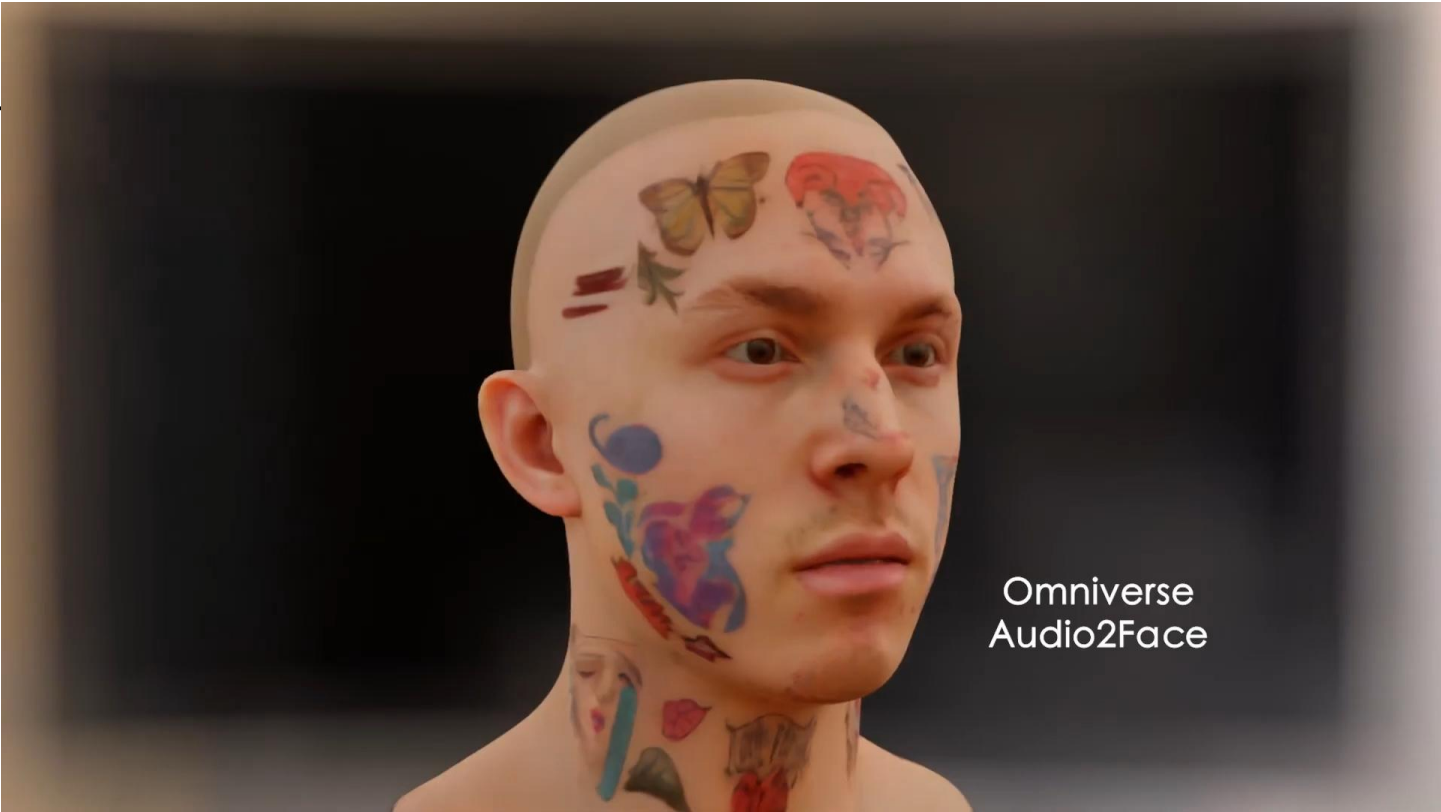


TransGS

TransGS

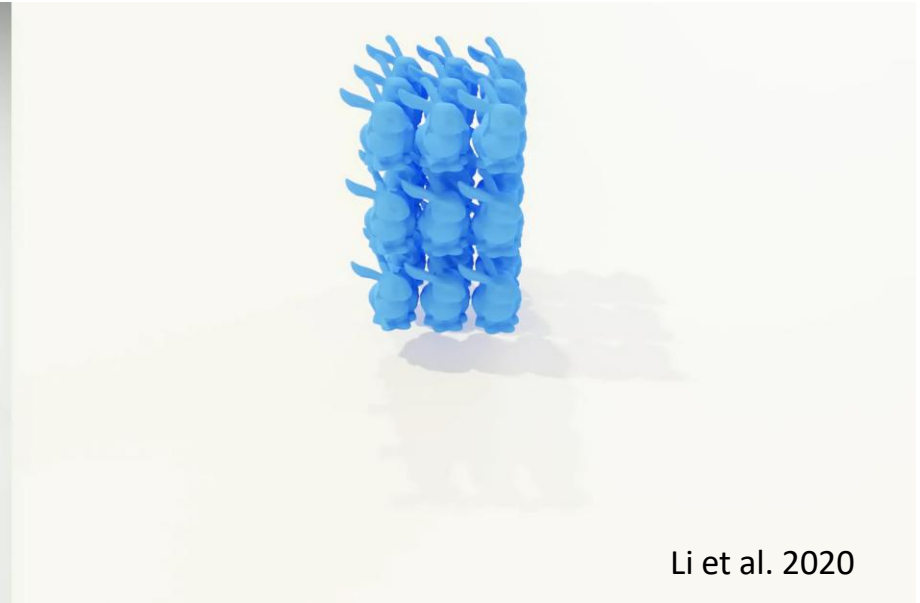
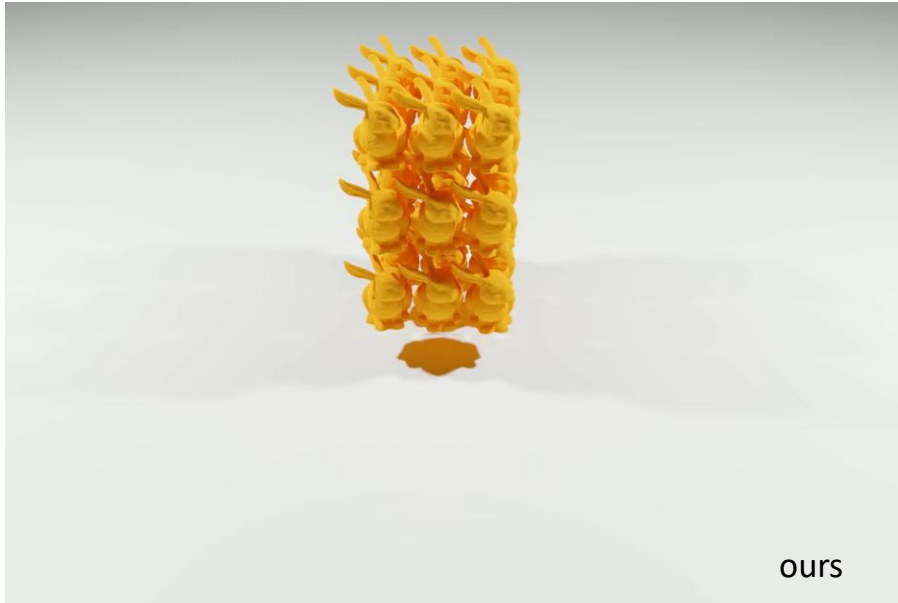
- Result

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TransGS

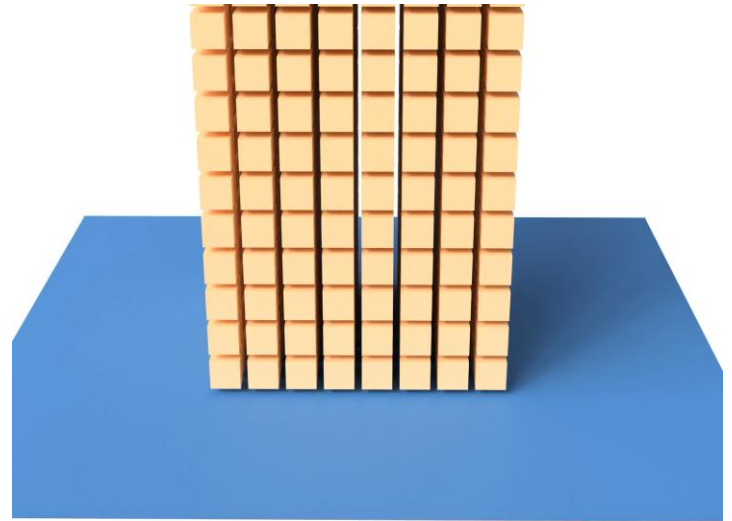
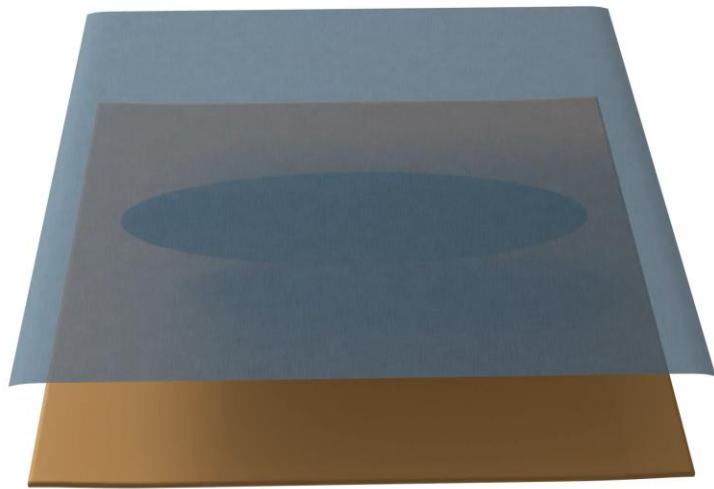
Falling Bunnies



2.2M tets, $dt = 0.01$

95x speedup!

Applications

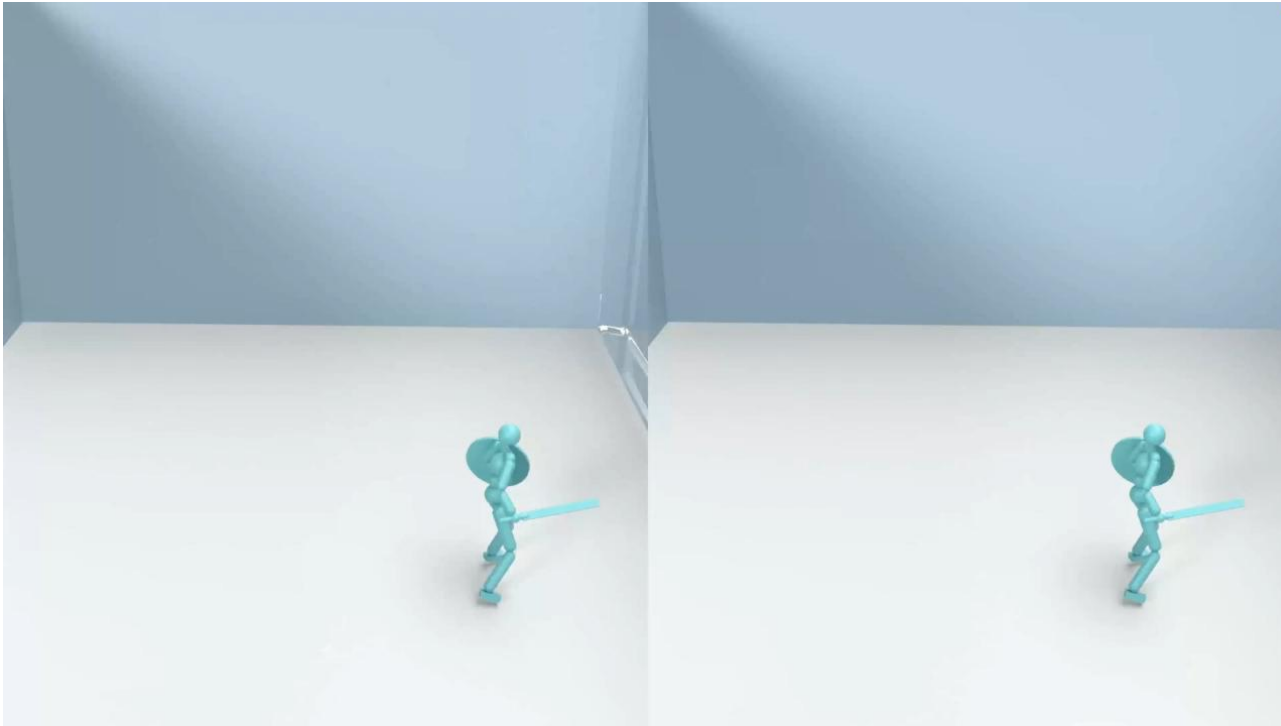


Character Control in Fluid

- Need to simulate the coupling of the body and fluid
- Optimizing the policy such that it achieves its objective while receiving external force from the fluid



Response to splash



Other Applications

In addition to films and computer games, applications of computer animation include

- AR/VR
- Creating data for machine learning
- Evaluation of buildings/environment

AR/VR applications



Preparing Training Data for Video Synthesis

Video synthesis is becoming very popular now, where videos are directly produced from user prompts.

I2V: Image2Video



In a sudden burst of energy, a cheerful man initiates squats mid-frame, transforming the serene office ambiance into a spontaneous workout moment. Dressed casually in a grey t-shirt and brown trousers, he sports a backpack, adding a quirky twist to his impromptu exercise. His smile remains intact, radiating positivity as he engages in fitness amidst the corporate setting. The camera captures this unexpected action in a medium full shot, emphasizing the contrast between his dynamic movement and the static background of



Initiate a slow camera pull-back from the initial framing to a full body shot. During the zoom-out, transition to show the same man grasping a barbell, tensing his muscles as if about to begin an upward lift. Keep the person recognizable.

T2V: Text2Video



Prompt: A sepia-toned vintage photo shows a whimsical dog bicycle race. Several dogs wear goggles and tiny cycling outfits, pedaling miniature bikes on a dusty road. Their determined expressions and blurred motion highlight the action. Spectators in period clothing line the sides, enhancing nostalgia. The image is grainy and softly lit, emphasizing warm tones and rustic charm. 'Bicycle Race' captures this lively scene in a medium shot, balancing racers and crowd.

I2V: Image2Video



3D Render



VideoGen

Prompt: Elegant Fashion Walk: A statuesque model, embodying poise and grace, strides confidently down the runway in a chic beige, longline coat. The coat cinches at the waist with a sleek belt, accentuating her slender silhouette. Her high heels click against the floor, adding rhythm to her purposeful gait. Bathed in soft, natural light from overhead skylights, she moves through a pristine, well-lit interior space. The video captures her dynamic movement, focusing on the fluidity of the fabric and the interplay of light and shadow as she walks. This is a full-length, moving portrait, shot from a side angle to showcase the model's elegant stride and the coat's refined design.



A2V: Audio2Video



Others: <https://omnihuman-lab.github.io/>

Preparing Training Data for Machine Learning

Applications like self-driving car require a lot of training data

- People start to use data created by computer animation for the training
 - Some researchers used GTA-5 animation for training their self-driving system
- Same concept applies to human motion tracking, facial expression tracking, etc.



Sim2Real



Motion Tracking

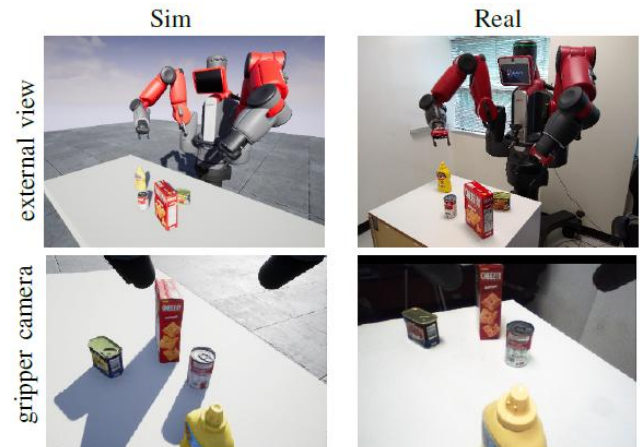
BeyondMimic

From Motion Tracking to Versatile Humanoid
Control via Guided Diffusion

Qiyuan Liao*, Takara Truong*, Xiaoyu Huang*, Guy Tevet, Koushil Sreenath†, C. Karen Liu†

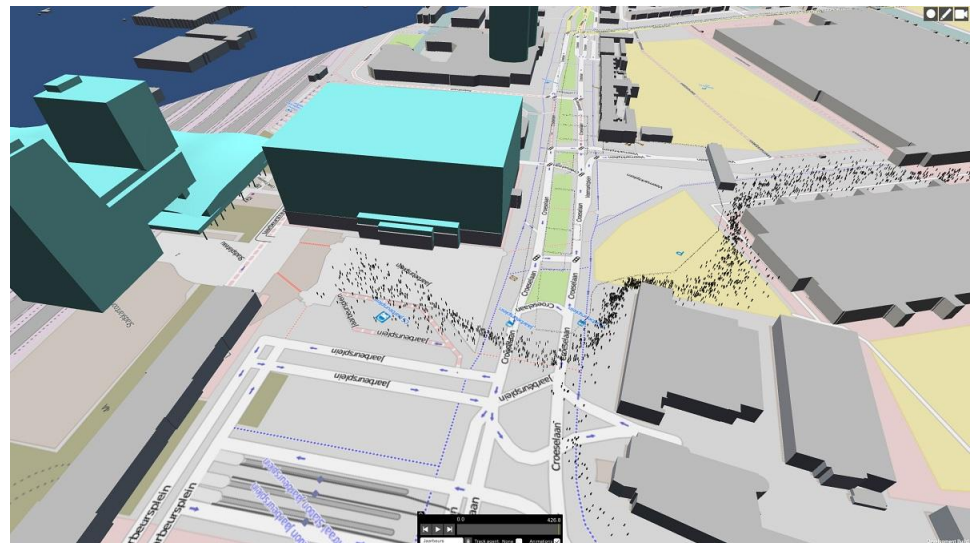
Sim2Real

- Machine learning is also used a lot for robot control
- Controlling the robot to collect data is time consuming
- That can also cause damage to the robot
- Instead, people simulate the robot in virtual environments and collect the data
- These are applied for the real robot for the control
- Usually there is a gap between simulation and the real robots – people use domain transfer techniques for filling in the gap.



Building/Environment Design

- The flow of people is an important factor when designing buildings or facilities like airports
- So that we know if the building is comfortable to live, or people can evacuate well in case of fire etc.
- Crowd simulation is used for such purposes.



Course Outline

- **30 Lectures**
 - Roughly half about character animation, half about physical simulation
- **6 tutorials**
 - Related to the assignments
- **4 Assessed Practicals (Python)**
 - 2 for character animation
 - 2 for physical simulation
- **Assessment**
 - Examination (50%)
 - Practical assignments (50%)

Q&A

